Approach B

Final methodology | Prefix-wise physical executability and hypothetical target restoration

Latent Causal Infeasibility for Long-Horizon TAMP

31 October 2026

FINAL RESEARCH DESIGN / IMPLEMENTATION STATUS
This document freezes the *agreed intended Approach B methodology*. Its prefix-wise X/Y factor graph and travel-aware objective are a **final specification to implement and evaluate**, **not** a claim that the existing P17 production LB model already implements them. P17 LB remains the frozen, code-verified baseline. This document contains Approach B only; it makes no decisions about Approach D.

Evidence basis. The 29 October 2026 methodology, Methodology v1.8, verified P17 B source and reports 0094/0097/0099, and the new `bd_core.py` / `bd_2d.py` protocol. The separate historical 0106 document prescribes a legacy-LB integration run, **not** this new X/Y training. The referenced private P17 0045 contract was not independently available for inspection.

Core question. Given a 2D robot scene and a **proposed ordered sequence of obstacle relocations to specified destinations**, can a model jointly estimate (i) whether the robot can **physically execute** each corrective prefix and (ii) whether the same prefix would **hypothetically clear the original target task**? Can it select a *successful*, *short*, *low-travel* correction more reliably than an uncoupled predictor or the existing direct-success LB baseline?

Definitive design choices. Two binary *outcome types* only, X and Y; one named quadratic X–Y coupling; F2 ordered-prefix and F3 hypothetical-scene unary scorers; F4 omission **queries** using the same scorers; F1 **analytic** endpoint screening; no generic F3/F4 pair functions, unrestricted F5 coupling or higher-order residual. Use one **root-deduplicated, masked joint Boltzmann NLL**, plus a travel-regression auxiliary, with deterministic action-count and displacement costs at selection. Budget-exhausted **attempted** queries equal binary 0; **unattempted** queries contribute no label.

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1 Scientific scope and frozen decisions

1.1 Problem and embodiment

The target object T has an original position and a specified target goal. Other movable objects may obstruct the intended target manipulation. A *corrective action* $a_t = (o_t, d_t)$ means physically relocating obstacle o_t to a specified legal destination d_t. A candidate repair is

$$r = (a_1, \dots, a_m), \qquad r_{1:t} = (a_1, \dots, a_t), \qquad 0 \le m \le 6.$$

The original target manipulation is attempted **after** the chosen corrective prefix. A root contains the initial object scene S_0, the initial robot configuration q_0, original target task and goal, fixed environment geometry, and a finite catalogue of proposed sequences. No new objects are inserted in counterfactuals.

The 2D physical teacher is the project's **articulated planar 3R arm** on the declared 8-by-8 tabletop lattice, with finite-width robot links and gripper, object geometry, supported side grasps, collision checking, attachment and release, and continuous robot configurations. Grid cells describe *resting placements*; the joint and carried-object paths are continuous. A proposal is not executable merely because its source and destination cells are empty.

Each relocation begins from the *actual terminal robot state of the preceding relocation*. No home reset between $A \to B \to C$ is permitted. The offline teacher may search grasps and continuous motions; the learned model does **not** generate IK, paths or gripper controls online.

Bounded challenge domain. The new dataset protocol declares a maximum of **seven total actions**: up to six corrective relocations and one final target action. The benchmark focuses on $K = 0, 1, 2, 3, 4$ **verified necessary** correction levels, with up to two redundant movements. Requested construction K is not automatically the verified K. Proposed destinations, correction count and sequence order remain explicit conditioning inputs; they are not predicted variables.

1.2 What the model claims and does not claim

Approach B learns **budget-policy-relative outcome probabilities for proposed high-level sequences**. In particular, it seeks to distinguish physical execution failure from target-restoration failure, detect order sensitivity and prefer economical repairs. It does **not** learn an object-wise binary blocker mask, identify a causal DAG, guarantee global physical impossibility or output a continuous plan. Learned quadratic factors are *statistical dependencies of controlled outcomes*; their causal interpretation comes from the independently defined offline interventions and teacher, not from an undirected factor edge.

A useful relocation may be insufficient to make the target immediately feasible: for three direct blockers, $Y_0 = Y_1 = Y_2 = 0$ and $Y_3 = 1$ is entirely expected. **Do not impose $Y_t > Y_{t-1}$ after each action**. In the intended direct-blocker scenes, a correctly chosen successful sequence typically reduces the *physical set of obstructions* as it progresses; the current X/Y outcome graph neither explicitly counts that set nor imposes a per-action decrement. Wrong destinations, nuisances and redundant actions remain valid negative/positive controls as generated.

2 Outcome variables: X and Y, precisely

2.1 Counterfactual scene state

For any candidate prefix $q = r_{1:t}$, construct a **hypothetical edited scene** S^q by applying the specified object destinations in order to S_0; the final object poses are deterministic. This edit does **not** assert that the arm could have performed those moves. Fixed geometry, the target definition and every nonedited object are unchanged. For target-only probes, use one **frozen target-reference robot state** q_{ref} (normally the original specified start q_0), a free gripper, no attachment, and the same target/goal and teacher budget for every matched edit.

Two ordered prefixes with **identical final object poses**, identical target definition and the same q_{ref} share *one physical Y query*, even if their action orders differ. They may have **different X queries**. This distinction is central to detecting order effects without inventing target-scene differences.

2.2 The two binary outcome types

For every unique **ordered physical prefix query** $q = r_{1:t}$,

$$X_q \in \{0,1\}: \text{ the proposed corrective prefix is physically executable under the frozen teacher protocol.}$$

The query includes S_0, q_0, the ordered object/destination list, allowed grasp/motion family and declared finite query budget. It requires valid approach, grasp, attached transport, placement, release and safe retreat for every relocation, with terminal joint state carried into the next action.

For every unique **edited-scene target query** $s = S^q$,

$$Y_s \in \{0,1\}: \text{ T succeeds in the hypothetically edited scene from fixed } q_{\text{ref}}.$$

The target-only label is generated even when physically relocating the obstacles is not executable; this is why $(X, Y) = (0, 1)$ is meaningful. Y_0 is the original target query with an empty edit, and for the zero-move repair $X_\emptyset = 1$ by definition if there are no corrective moves and the target-reference state is already available.

2.2.1 Essential connector condition for interpreting joint success

The pair $X_q = 1, Y_{S^q} = 1$ alone does **not** imply a physically executable *full* repair if X only verifies relocation and ends at an arbitrary robot configuration. To make the X/Y success event compositional, **the finalized X query includes a validated, collision-free connection from the terminal relocation state to the frozen target-reference start q_{ref} in the edited scene**, with the gripper free and no attachment. Then a valid X witness can be concatenated with the separately validated Y target trajectory. This terminal connector is tested **only when a prefix is considered as a potential stopping point**; it is *not* a home reset between successive corrective actions. Connector cost enters travel J.

This is a **new teacher-interface requirement** for the finalized method. The old new-2D generator's `verify_sequence()` invokes target feasibility from the **achieved** robot state and does not furnish these X-with-connector / fixed-reference-Y pairs. Its existing labels must not be silently relabeled as the finalized X/Y pairs. A separately validated full sequence remains an independent external execution check, not a third learned variable.

2.3 Four valid joint outcomes

X_q	Y_{S^q}	Interpretation	Complete repair + target composable?
0	0	Cannot complete the corrective prefix within budget; target also fails in edited scene	No
0	1	Hypothetical edit clears T, but the robot cannot execute and connect that prefix within budget	No

X_q	Y_{S^q}	Interpretation	Complete repair + target composable?
1	0	Correction and connector succeed, but T remains infeasible from the fixed target start	No
1	1	Correction and connector succeed; T succeeds from the same target-reference state	**Yes, under the stated teacher protocol**

No separate learned Z_R is necessary. The desired event for a complete candidate r is $X_r = 1$ **and** $Y_{S^r} = 1$.

Operational binary labels. Every actually attempted query receives 1 only upon full verification within its pre-declared budget; otherwise it receives 0, including timeout, capped search, failed contact mode or exhausted planning attempts. Keep `negative_kind`, failed phase, attempt count and budget **as metadata, not as a third training class**. A query **never attempted** is absent from the observed set, not an observed zero. The operational outcome is deliberately **not** a proof of global infeasibility.

3 Final factor-graph structure

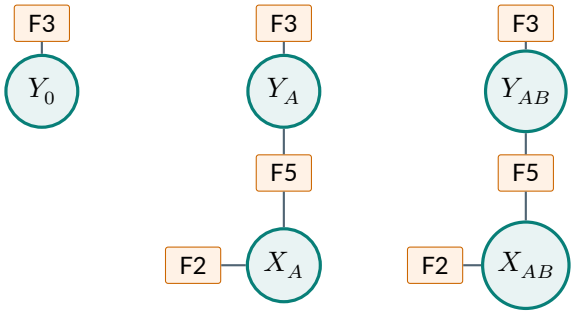
3.1 Frozen simplified factor families

Family	Final purpose	Status in finalized B
F1: static endpoint	Check support, occupancy, pair overlap, immovable structure and legal destination constraints	**Analytic hard screen only**; no trained static BCE or outcome coupling
F2: ordered sequence	Encode actual action identities, destinations, robot start and ideal object prefixes; produce **X unary logit** a_q	**Retained**, prefix-predecessor model; no GRU required
F3: scene intervention	Encode counterfactual S^q from fixed q_{ref}; produce **Y unary logit** b_s	**Retained** as unary; old scene–scene learned pair potentials removed
F4: omission	Query the same X/Y model on a **new, recomputed** sequence with one action removed	**Retained as queries**, not a separate learned repair–omission pair function
F5: coupling	Couple the physical-outcome X node with its corresponding hypothetical-scene Y node	**Only the named quadratic X–Y pair**, no generic cross-outcome or higher-order network

The older LB experiments favored dropping generic coupling from deployment; the revised X–Y interaction is a *new, controlled hypothesis* and is **not** supported merely by those older experiments. F1's old learned static logits and F3/F4 extra pair terms are not reintroduced. The travel head is separate from these feasibility factor families.

3.2 Directed normalization for a root-consistent undirected factor graph

A root may contain many candidate prefixes. Let $\mathcal{Q}$ be the set of **distinct ordered-prefix X queries** and $\mathcal{S}$ the set of **distinct hypothetical scene Y queries**. Let $s(q)$ be the unique scene-outcome key corresponding to prefix q. The same Y node can be shared by several X queries if they have the same hypothetical final object poses.



Illustrative prefixes A and $A \to B$. Circles are outcome variables; squares are learned factors. Omission queries reuse these nodes/functions; no separate F4 pair potential.

The figure illustrates the local factors; it is **not** a causal directed acyclic graph. The chronological $A \to B$ order is a conditioning property of the F2 neural input, not an arrow between random variables. For two orderings $A \to B$ and $B \to A$ with equal endpoints, draw two different X circles coupled to **the same** Y_{AB} circle.

A naïve globally normalized $\exp[-\sum_q E_q(X_q, Y_{s(q)})/T]$ would allow the *number of listed candidate sequences* to change the marginal probability of a scene outcome Y_s. That is undesirable. Therefore use a **root-consistent, normalized scene-first factorization**:

$$\boxed{P_\theta(y_{\mathcal{S}}, x_{\mathcal{Q}} \mid c, \mathcal{Q}) = P_\theta^Y(y_{\mathcal{S}} \mid c) \prod_{q \in \mathcal{Q}} P_\theta^X(x_q \mid y_{s(q)}, c, q).}$$

This is a probability factorization; the corresponding factor graph still contains undirected unary and X–Y pair factors. The probabilistic factorization **does not claim that hypothetical Y physically causes X**. It enforces root-consistent scene marginals and prevents candidate-list artifacts. No higher-order or X–X/Y–Y pair potentials are included in this finalized primary model.

4 Exact Boltzmann energy and normalization

4.1 Shared neural coefficients

Let $v_s = f_{\text{scene}}(S^s, q_{\text{ref}}, T, g_T)$ be the shared scene representation. Let $g_q = f_{\text{order}}(S_0, q_0, a_1, \dots, a_t)$ be the shared prefix-order representation. Define the trainable logits (temperature $\tau > 0$, frozen initially to 1):

$$\begin{aligned} b_s &= h_Y(v_s), && \text{hypothetical target unary (F3),} \\ a_q &= h_X(g_q), && \text{physical prefix unary (F2),} \\ w_q &= h_{XY}(g_q, v_{s(q)}), && \text{specific quadratic coupling (F5).} \end{aligned}$$

The X–Y pair energy for one query is **exactly**

$$\boxed{E_q(x, y) = -a_q x - b_{s(q)} y - w_q x y, \qquad x, y \in \{0, 1\}.}$$

Here b_s is **a neural coefficient for the Y outcome, not an object-blocker variable**. The xy coefficient is learned, not assigned from a collision heuristic. It represents a context-dependent *log-odds interaction*. Unlike the previous achieved-state-Y proposal, **all four X/Y combinations are permitted**, so the interaction is not algebraically redundant. The energy does not itself count blockers or guarantee that the target succeeds after each action.

4.2 Root-consistent normalization (the authoritative probability model)

Use a scene Bernoulli potential

$$P_\theta^Y(Y_s = y) = \frac{\exp(b_s y/\tau)}{1 + \exp(b_s/\tau)}$$

and a scene-conditional physical potential

$$P_\theta^X(X_q = x \mid Y_{s(q)} = y) = \frac{\exp\big((a_q + w_q y)x/\tau\big)}{1 + \exp\big((a_q + w_q y)/\tau\big)}.$$

These are two normalized Boltzmann factor families with **one quadratic interaction**, not two independently optimized loss heads. Their product is a valid normalized joint distribution over all distinct root query outcomes. Equivalently, its **root-normalized energy** is

$$\boxed{\begin{aligned} E_\theta^{\text{root}}(x,y) = &\sum_{s\in\mathcal{S}} \left[-b_s y_s + \tau \log(1 + e^{b_s/\tau})\right] \\ &+ \sum_{q\in\mathcal{Q}} \left[-(a_q + w_q y_{s(q)})x_q + \tau \log\left(1 + e^{(a_q + w_q y_{s(q)})/\tau}\right)\right]. \end{aligned}}$$

The Y-dependent second normalizer is essential. It prevents X–Y coupling from changing the scene-only marginal when additional physical candidates are listed. The normalized energy has $P_\theta(x,y) = \exp[-E_\theta^{\text{root}}(x,y)/\tau]$ and partition function 1 **because the normalizers are already included**. Alternatively, omit those normalizers and explicitly normalize the appropriate blocks; never use the raw quadratic alone as a normalized root model.

The fixed scene probabilities and conditional X probabilities are

$$P(Y_s = 1) = \sigma(b_s/\tau), \qquad P(X_q = 1 \mid Y_s = y) = \sigma((a_q + w_q y)/\tau).$$

The required **joint correction-and-target success probability** is

$$\boxed{p_{\text{succ}}(q) = P(X_q = 1, Y_{s(q)} = 1) = \sigma(b_{s(q)}/\tau)\,\sigma((a_q + w_q)/\tau).}$$

For diagnostics, $P(X_q = 1)$ marginalizes Y_s rather than substituting a teacher-revealed Y_s value at inference. The pair's cross-log-odds ratio is w_q/τ. A nonzero coupling can model residual dependence beyond separate scene and order unary logits, but **a learned coefficient is not proof of a causal edge**. Whether this coupling improves held-out predictions or repair decisions is an explicit experiment.

4.3 F2 ordering: action-prefix interactions, not X/Y coupling

Each proposed action receives a token h_t from the object identity, source/destination geometry, original robot state, current ideal **object-only** prefix, and shared scene representation. B's existing predecessor-aware construction computes an action summary from its predecessors:

$$g_q \;=\; \text{Pool}_{t\le|q|}\ \text{Mix}_\theta(h_t, \text{Pool}_{u<t}\,\text{Pair}_\theta(h_u, h_t))\,.$$

This equation captures the existing P17 **prefix-predecessor** design; exact learned MLP definitions are implementation details. Crucially, the neural Pair(h_u, h_t) inside **F2** is **not** the quadratic Bernoulli $w_q X_q Y_s$ inside **F5**. F2 produces a_q and order-dependent context for w_q; F5 couples X/Y outcomes.

Two proposed permutations can have $S^{AB} = S^{BA}$, hence **the same Y**, while $g_{AB} \neq g_{BA}$, hence different X predictions. Robot joint dynamics are not simulated by the encoder; full offline teacher verification supplies X supervision. An ideal prefix scene must not be described as a witnessed robot state.

4.4 F4 omissions and repeated physical queries

For $q = (a_1, \dots, a_m)$, define $q^{-t} = (a_1, \dots, a_{t-1}, a_{t+1}, \dots, a_m)$. Reconstruct the remaining actions' ideal prefixes **from the original scene**, then evaluate the **same** X/Y scorer on q^{-t}. Do not infer a label by simply removing one segment from an original executed trace. The destination and order of every retained action stay unchanged.

Two omission queries that generate the same ordered prefix share one X variable. Two different orders with identical edited object poses share one Y variable. The no-op original target Y_0 is also shared. Do not duplicate factor instances just because a query is mentioned under several candidate roles. No distinct F4 pair MLP is trained.

5 Learning from the offline teacher

5.1 A single recorded root

For each scene root, store its initial scene, q_0, q_{ref}, geometry, target and target goal, exact proposed candidate catalogue, and the complete declared budget profile. For **every actual teacher query**, store a canonical key, attempted flag, binary y, failure metadata, configuration/robot profile hash, and any successful motion witness. Preserve source/target object identities and destination cells.

For a proposed prefix q, the required teacher calls are logically **two independent experiments**:

1. **X query**: from S_0, q_0, physically execute the full ordered prefix, preserving the achieved joint state between relocations, then connect from its final state to q_{ref} in the resulting scene. Stop at the first unsuccessful step under the declared teacher policy. Log the first failing action/connector, successful prefix witnesses and complete travel when available.
2. **Y query**: from the *hypothetically edited* S^q at the same fixed q_{ref}, independently execute T with the declared target success/retreat condition. Y is queried **even when X=0**, because that is the informative $(0, 1)$ case. Cache identical edited-scene queries exactly once per root.

For an observed X query, a failed attempt is $X = 0$ without a third unknown class. If an optional intermediate Y is not executed, it has *no observed target label*; this must not be converted to Y=0. In the finalized collection, independently query Y for every selected prefix, subject to a predeclared budget, and label every **attempted** query binary.

Physical-query caution. The current `bd_2d.py` constructor uses `p16_common.verify_sequence()` with a greedy selection of one successful motion mode per action and then tests T from the **achieved** state. The P17 `p17_verify.verify_existential()` teacher searches multiple motion realizations. They are different policies. The X/Y dataset must choose one *fixed*, *versioned* teacher policy and apply it consistently; do not mix or silently translate old complete-sequence labels into the new variables. The first run of historical LB on the existing dataset remains a **separate frozen baseline experiment**, not evidence that X/Y has been trained.

5.2 Existing data versus required extensions

Observation	Existing new-2D `BD_GEN_V1` record	Final X/Y need
Original target from q_0	Yes: `labels.original`	Reuse only if q_{ref}, teacher and task scope match
Full ordered repair + T	Yes: `labels.B`, `failed_step`, successful witness	Retain as whole-execution baseline; **not** substitute for X/Y
Fixed-robot scene edits	Selected `labels.D` edits	Add exact target-only Y for **every selected ordered prefix**
Executable prefix + terminal connector to q_{ref}	Not systematically recorded	Add independently evaluated X prefix records

Observation	Existing new-2D `BD_GEN_V1` record	Final X/Y need
Travel of successful complete trajectories	Recorded for some successful B queries	Record matched X connector + T travel for full-success candidate
Same-endpoint reversed-order X contrast	Selected proposals, no guaranteed decided pair	Require real physical attempts in both orders

Do not overwrite old data. Use immutable source records and a versioned **Approach-B-X/Y sidecar** with explicit parent root/query IDs, label/teacher protocol hashes and new outcomes. This also permits a faithful direct LB baseline on exactly the original recorded labels.

5.3 Missingness, duplicates and the joint observed-data loss

Let $\mathcal{O}_Y \subseteq \mathcal{S}$ and $\mathcal{O}_X \subseteq \mathcal{Q}$ index **attempted and recorded** query outcomes. Because some labels may not have been collected, optimize the **marginal probability of the observed root evidence**:

$$\mathcal{L}_{\text{joint}}(\theta) = -\frac{1}{N_{\text{roots}}} \sum_{c} \log P_\theta(D_c \mid c),$$

$$P_\theta(D_c \mid c) = \sum_{y_{\mathcal{S}}} \left[\prod_{s \in \mathcal{S}} P_\theta^Y(y_s \mid c) \right] \left[\prod_{s \in \mathcal{O}_Y} \mathbf{1}(y_s = \bar{y}_s) \right] \times \prod_{q \in \mathcal{Q}} \left[\sum_{x_q \in A_q} P_\theta^X(x_q \mid y_{s(q)}, c, q) \right],$$

where $A_q = \{\bar{x}_q\}$ for an observed X and $A_q = \{0, 1\}$ otherwise. There is **one joint feasibility likelihood**, with no separately weighted physical or target BCE losses. It marginalizes unobserved variables exactly. Root averaging avoids a root with many proposed sequences dominating training. The same physical outcome recorded twice is not counted twice. With only the retained X–Y pairs and independent Y unaries, this summation factorizes by unique scene Y, so exact computation need not enumerate $2^{|\mathcal{S}|+|\mathcal{Q}|}$ configurations.

Important scientific constraint. Every term in the full label product is a *distinct physical query*. Independent X queries are conditionally independent given their shared Y in this deliberately lean model. This is a probabilistic approximation, not proof that robot executions in the same scene are independent. Shared sequence-prefix evidence is communicated through input features; do not introduce extra X–X edges without a separately tested reason.

6 Travel objective and corrective selection

6.1 Cost variables

For any candidate r, define the following separately:

- $m(r)$: **exact** number of corrective relocations (not a prediction).
- $C_d(r) = \sum_{a_t=(o_t,d_t)} \|d_t - p_{o_t}^{\text{initial}}\|/L$: **exact** summed obstacle displacement, using one pinned normalizing length L and the actual proposed destinations.
- $J(w)$: measured **joint-space travel** for a successful, complete witness. Use one published metric, e.g. $J = \sum_k \|q_{k+1} - q_k\|_1$, consistently over approach, carry, release, inter-action transitions, terminal target-reference connector and final target, with an unwrapped/consistent joint-angle convention and a frozen normalization. Gripper transitions and time may be logged separately but must not silently redefine J.
- $\hat{J}_\theta(r) > 0$: a **learned cost head** from the candidate's sequence features, supervised only when a matched complete witness exists. It is not a force, kinetic energy or motion trajectory.

The observed travel is **teacher-policy-specific** and, for a first complete witness, not necessarily the shortest achievable robot path. Distinguish joint travel from object displacement: a short object motion can require large robot travel.

6.2 Training loss and hard eligibility

Define a robust, positive-cost regression term, for example

$$\mathcal{L}_J = \frac{1}{N_J} \sum_{w \in \mathcal{W}_{\text{complete}}} \text{Huber}\Big(\log(1 + \hat{J}_\theta(r_w)), \log(1 + J(w))\Big) .$$

The **final training objective** is

$$\boxed{\mathcal{L}_B = \mathcal{L}_{\text{joint}} + \lambda_J \mathcal{L}_J .}$$

One joint probability loss plus one optional, justified travel auxiliary. Set the primary travel coefficient on validation data; compare against $\lambda_J = 0$. A failed sequence does not receive a fabricated large travel label. F1 static/object legality is evaluated using analytic geometry/support checks **before scoring**, not an independent learned penalty and not a trained BCE term. Static-only penalties do not modify $P(X, Y \mid r)$ by being added as a constant to all of its outcome energies.

6.3 Inference and proposal ranking

Given the **same frozen finite proposal catalogue** for every compared method:

1. Apply analytic **sound-invalid** placement/support screens (F1); remove only candidates ruled out at the declared scope. Keep the zero-move candidate.
2. Compute all unique Y scene logits, all X ordered-prefix logits and X–Y coupling terms without teacher/privileged inputs. Marginalize, do **not** condition on the actual test outcome of Y.
3. For each complete candidate r, compute $p_{\text{succ}}(r)$ from the joint model. If the original task passes the no-op rule, do not invent obstacle movements.
4. Admit candidates above a **validation-frozen feasibility threshold** η; if none qualify, abstain or invoke a **separately declared** safe baseline. Never treat a high learned probability as overriding a sound analytic certificate.
5. Rank admitted candidates using a validation-frozen cost/risk rule:

$$\boxed{C(r) = -\log p_{\text{succ}}(r) + \alpha\, m(r) + \beta\, C_d(r) + \gamma\, \hat{J}_\theta(r).}$$

Then select $r^* = \arg\min_{r \in \mathcal{C}_{\text{admit}}} C(r)$; deterministic signature breaks ties. Coefficients α, β, γ are decision hyperparameters, **not** automatically the same as training weight λ_J. An optional lexicographic length-first selector is a preregistered ablation, not a silent change in the method.

After selection, the executable movement must come from the independent, versioned physical motion provider and its verified trajectory/connector evidence. B does not synthesize continuous motion or generate new destination coordinates. A perfect learned score cannot select a repair absent from the proposal catalogue. No online teacher label is revealed to the scorer.

7 Worked example: three direct target blockers

Let A, B and C be three objects obstructing the target sweep. Suppose the proposer supplies valid supported destinations p_A, p_B, p_C. Consider the *illustrative* (not measured) target-only outcomes:

Hypothetical prefix scene	Original target Y	Meaning
S_0	0	T blocked initially
S^A	0	B and C still obstruct
$S^{A,B}$	0	C still obstructs
$S^{A,B,C}$	1	Target succeeds after all three edits

Now suppose the physical arm cannot execute $A \to B \to C$ under the declared budget because its arm pose after A blocks the B relocation, but $B \to A \to C$ is executable. Both orders end in the same edited object scene, so

$$Y_{ABC} = Y_{BAC} = 1, \qquad X_{ABC} = 0, \quad X_{BAC} = 1.$$

This is **precisely** the benefit of separating X from Y: one edited-scene target outcome is shared, while order-dependent physical feasibility differs.

The learned coefficients may be illustrative values such as $(a_q, b_s, w_q) = (-1.0, 2.0, 0.5)$ for one order and $(0.5, 2.0, 1.0)$ for the other; these are **not** ground-truth probabilities or experimentally measured effects. X and Y form a joint prediction using the same context-dependent energy family. If a successful alternative sends A farther but reduces measured joint travel, cost-aware selection can prefer the shorter **robot** motion, subject to the other costs.

The example does **not** require Y to change after every relocation. The decreasing direct-blocker set is a geometric description of correct edits, not a sequence of distinct positive Y labels.

8 Implementation contract for the new 2D experiment

8.1 Minimal code boundaries

Keep the frozen historical B implementation and archived evidence unchanged. The original LB model is `abc7_models.ModelBC7(kind="B",couple="none",order="pred")`, trained by keyed unique-query NLL with no trained static/travel auxiliaries. It predicts *whole-repair outcome queries*, **not** X/Y. Do not rename the new X/Y implementation LB, overwrite its checkpoint or report old LB numbers as new results.

Implement the new B as a **versioned wrapper or B-specific module** reusing the existing scene encoder and predecessor-aware action-token code. Add only: (i) X prefix logits, (ii) Y hypothetical-scene logits, (iii) the one pairwise XY coupling head, (iv) the root-consistent exact likelihood with shared Y query keys, (v) the travel readout, and (vi) a minimal data-sidecar reader. Do not rewrite the dataset generator, motion-planning code, Approach D modules, local/global similarity or swept-volume retrieval as part of B learning.

The new sequence representation must support **up to six obstacle relocations**, without silently truncating historical four-step layouts. Do not retain `MAX_ACTIVE=9` as a hard cap on the new X/Y query graph; compute the lean normalized distribution by query blocks. Include explicit q-reference and queried-prefix canonicalization tests.

8.2 Training and evaluation protocol

Partition. Use the existing `BD_GEN_V1` root/family grouped IID split for the primary experiment, with no scene descendants, sibling edits, reversed-order twins or renderer variations crossing splits. K-stratified and held-out geometry/obstacle-count tests are secondary when their actual coverage suffices. No historical P13–P17 roots may enter new-dataset training. Freeze random seeds, data hash, model version, motion-teacher version, caps and eligibility policies before final testing.

Primary learning comparison (matched supervision and candidate catalogue):

Arm	What is trained	Why present
LB (frozen reference)	Original direct sequence outcome; original task-label protocol	Historical baseline; its outcomes are **not** magically X/Y
`XY-uncoupled`	Same shared F2/F3 representations, $w_q = 0$, joint observed-data NLL	Tests whether separate physical/target prediction is sufficient
`XY-quadratic` (primary)	Same representations with context-dependent w_q; root-exact joint NLL	Tests value of the single retained coupling
`XY-quadratic-noJ`	Same coupling, $\lambda_J = 0$	Tests travel auxiliary's effect on feasibility and selection

The two XY arms must train on **identical observed X/Y sidecar labels** and use matched parameter/compute controls if the coupling head changes capacity substantially. Keep the complete task outcome evaluation separate from label acquisition: use a **held-out physical replay/verification** for selected proposals and disclose provider cost. Compare an analytic geometry/shortest-candidate rule on the same catalogue. A no-op-first and a verify-first reference are required where operationally possible.

For all arms report: root-level joint NLL (where defined), Brier and reliability/calibration of $P(X = 1)$, $P(Y = 1)$ and p_{succ}, held-out full-repair outcomes, correct failure-stage localization, original-feasible false-repair rate, repair-selection success, optimality *within the finite verified catalogue*, chosen $m(r)$, C_d and measured J, inference CPU and offline teacher labeling CPU. Report matched same-endpoint reversed-order pairs. Stratify by actual verified K0–K4, sequence length, redundant actions, displacement bin, distractor count, original robot start and mechanism where sample counts permit. Give number of independent roots and paired uncertainty intervals, not just per-query pooled accuracy.

Acceptance checks before claiming final-model benefit:

1. **Semantics:** Y is invariant to the order of identical endpoint edits; X can differ by order; none of the hypothetical target labels is contaminated by achieved-q target outcomes.
2. **Physics:** each positive X has a replayed relocation-plus-connector witness preserving exact achieved joint states; each positive Y has an independent target-only witness from the pinned reference state.
3. **Binary contract:** all attempted teacher failures within budget are 0; no silently introduced UNKNOWN *labels*; unattempted queries are never fabricated zeros.
4. **Math:** normalized four-state X/Y microcases agree with brute-force enumeration; shared Y marginals are invariant to appending extra X proposals; analytic gradients and marginal likelihood agree with reference microcases.
5. **Identity:** duplicate aliases and omissions are deduplicated; the zero-move case is correct; all six-step sequences survive feature encoding.
6. **Coupling:** XY-quadratic must outperform XY-uncoupled in a declared meaningful metric before attributing value to the interaction; do not claim novelty from the mere presence of an xy term.
7. **Travel:** compare $\lambda_J > 0$ versus $\lambda_J = 0$ without sacrificing calibrated feasibility; cost is measured on the same physical protocol.
8. **End-to-end:** candidate selection is evaluated on independent scenes with a frozen provider and equal proposal availability; no hidden target-oracle results or training-witness trajectory leaks into inference.

If the XY-quadratic arm does not outperform its uncoupled control, **remove the learned XY coupling from the eventual deployed version** rather than declaring it necessary by design. If stagewise learning fails to improve on direct LB under matched evidence and compute, report that result and preserve LB as the simpler alternative. No single model family is presumed superior.

9 Limitations, validity and final methodological statement

- **Statistical vs causal:** The nonzero w_q coefficient is a joint outcome interaction, not an identified causal arrow from Y to X. The independently designed scene interventions and physical sequence contrasts are

what make causal statements testable.

- **Binary target signal:** Y need not become positive until the last necessary blocker is relocated. The model does not explicitly infer a decreasing blocker count and must not claim it does.
- **Teacher scope:** Binary zero means failure **under the declared budgeted teacher protocol**; it is not a universal impossibility certificate. The failed phase is preserved as metadata without changing the binary label.
- **Motion realizations:** X is a budgeted-existence statement under a fixed policy and its terminal connector requirement. It is not a low-level plan generator or guarantee under an arbitrary new controller.
- **Counterfactual domain:** Y probes rearranged object poses from fixed q_{ref}, not achieved robot states. X witnesses with connectors make positive XY pairs physically composable under the pinned interface. Generalizing to other robot starts requires fresh matched data.
- **Catalogues:** Learned success probabilities cannot rescue physically valid sequences missing from the candidate generator. Report candidate recall separately from classifier discrimination.
- **Travel:** Predicted travel is the cost of the sampled successful teacher-witness policy, not globally minimum joint travel; only observed complete witnesses supervise it.
- **Evidence:** Historical LB/FULLB P17 results justify the lean simplifications but do not demonstrate advantages of the new X/Y coupling. Actual K3/K4 coverage, order contrasts and composable successful X/Y pairs must be confirmed by the new collection.

FINAL APPROACH B IN ONE SENTENCE

A root-consistent, prefix-conditioned Boltzmann factor graph learns *two* operational binary outcome types – **X**, whether the ordered obstacle-relocation prefix and terminal target-reference connector can physically execute, and **Y**, whether the same hypothetical edited scene permits the original target task – using retained ordered-sequence and scene unaries, one testable quadratic X–Y coupling, omission queries, analytic static screening, and a separate learned joint-travel objective for cost-aware selection.